MILITARY SPECIFICATION

IGNITER ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Igniter Assembly, Part Number 11740321, for use in conjunction with the Rocket Motor, M119 in an aircrew emergency escape system .

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-R-50753	Rocket Motor, M119 Assembly

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-453	Inspection, Radiographic
DOD-STD-2101	Classification of Characteristics

2.1.2 *Other Government documents, drawings and specifications.* The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Ordnance Station, Indian Head, Maryland (Code Ident 14083)

11740321	Igniter Assembly

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article sample. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Materials. All materials, parts and assemblies shall be in accordance with the documents listed on Drawing 11740321.

3.3 Protective finish. Protective finish shall be in accordance with the applicable specification referenced on the drawings and shall apply to all surfaces regardless of configuration.

3.4 Age

3.4.1 Explosive components. All explosive components including but not limited to ignition powders and igniters shall be manufactured or recertified within 12 months of the delivery date.

3.4.2 Primers. Primers shall have been manufactured and certified within 24 months or recertified within 12 months of the original contract delivery date.

3.5 Leak. The sealed igniter assembly shall not exhibit a leak rate of greater than 1×10^{-5} cubic centimeters (cm^3) of air per second at a pressure differential of 1.0 ± 0.1 atmosphere (see 4.5.4.1).

3.6 Radiographic. Each igniter assembly shall be radiographically inspected. Interal configuration and condition shall be as specified on drawing 11740321 (See 4.5.4.2)

3.7 Functional. The igniter assembly shall not fail to fire.

3.8 Ballistics. The igniter assembly, loaded with acceptable explosive components, when fired in an M119 Rocket Motor shall ignite the rocket motor propellant within .015 seconds maximum at temperatures from minus 65 degrees to plus 160 degrees Fahrenheit (F).

3.9 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects. All components shall be free from cracks, splits, laminations, perforations, porosity or other metal defects.

3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter. There shall be no excess sealant on surfaces adjacent to the sealed parts.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 65° F to 95° F
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.4 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production.

4.4.1 Sample size. The sample size shall be as specified in the contract (see 6.2).

4.4.2 Inspection routine. When the igniter is manufactured as a subassembly of the M119 rocket motor, the complete unit shall be tested in accordance with MIL-R-50753. If the igniter is not manufactured as a subassembly, the first article inspection requirements shall be specified in the contract.

4.4.3 Failure. Failure of any igniter to pass the first article tests shall be cause for rejection of the first article sample.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examinations and test of 4.5.1 thru 4.5.4 conducted on samples selected as specified in 4.5.1. The contractor is responsible for assuring that all igniter assemblies offered for acceptance meet all the requirements of the applicable drawings and specifications. Items of inspection shall be serialized and inspection data traceable to a specific serial number.

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4.5.1 Sampling for quality conformance.

4.5.1.1 Lot. A lot shall consist of an identifiable group of igniter assemblies consisting of:

a. Components produced by a homogeneous process.
b. Primers from not more than one lot.
c. Ignition material from not more than one lot.
d. Ignition pellets from not more than one lot.

4.5.1.2 Sampling for examination. Designation of defects (critical, major and minor) shall be in accordance with DOD-STD-2101.

a. Critical and major defects: Each igniter assembly shall be examined for all critical and major defects.
b. Minor defects: Samples for examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.5.1.3 Sampling for non-destructive test. Each igniter assembly shall be subjected to the tests specified in 4.5.4

4.5.2 Examination.

4.5.2.1 Critical and major defects. Each igniter assembly shall be visually and radiographically examined for critical and major defects specifed in 4.5.1.2 and the applicable drawings. Any igniter assembly found to contain a critical or major defect shall be rejected.

4.5.2.2 Minor defects. Sample igniter assemblies selected in accordance with 4.5.1.2 shall be examined for minor defects in accordance with the applicable drawings. The acceptable quality level (AQL) shall be 2.5 defects per hundred units.

4.5.3 Production lot acceptance test firing. The igniter is manufactured as a subassembly of the M119 rocket motor. The complete unit shall be tested in accordance with MIL-R-50753. A separate product lot acceptance is not required.

4.5.4 Test methods.

4.5.4.1 Leak test. The igniter assembly shall be subjected to a 100 percent leak test. The igniter shall be subjected to a helium diffusion leak test as follows:

1. Place the igniter in a vacuum chamber and decrease chamber pressure to 1 inch Hg maximum for 25 minutes.
2. Break chamber vacuum pressure by back filling the chamber with helium and pressurizing to 15 psig for a minimum of 5 minutes.
3. Reduce chamber pressure to ambient conditions and place the igniter immediately into a helium atmosphere for 5 minutes.
4. Remove the igniter from the helium atmosphere and remove superficially entrapped helium from the igniter using dry nitrogen at 20 psig.
5. Place the igniter in the leak detector vacuum chamber.
6. Leak test the igniter within 20 minutes after removal from the pressurized chamber.
7. All igniter whose leakage rate is in excess of 1×10^{-5} standard cc/sec air shall be rejected.

Failure of any igniter assembly to comply with the requirements of 3.5 shall be cause for rejection of the igniter assembly.

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4.5.4.2 Radiographic inspection. Each igniter assembly shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level 1. The igniter assembly shall be x-rayed in one view normal to the longitudinal axis. Any deficiencies shall be cause for rejection of the igniter. Radiographs shall be capable of detecting defects and igniter components 0.03 inch in length or width whichever is smaller. If all components cannot be shown using standard radiographic techniques, neutron radiographic shall be employed to show these components. All radiographs shall be made available for inspection by a designated government representative.

5. PACKAGING

5.1 Packing and marking. The igniter assembly is a part of the M119 Rocket Motor Assembly and is not packaged separately. See packaging instructions for the M119 Rocket Motor Assembly in MIL-R-50753.

6. NOTES

6.1 Intended use. Igniter assemblies specified herein are intended for use as the ignition element for the M119 rocket motor in an aircrew emergency escape system.

6.2 Ordering data.

6.2.1 Acqusition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification
b. Inspection conditions, if other than as specified (4.3)
c. If first article is required and sample size, (3.1, 4.4, 6.3, 6.4)
d. Facility for performing inspection and lot acceptance test firings (4.4, 6.5)
e. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5 are applicable.

6.3 First article. When a first article inspection is required, the contract shall specify first article sample. The first article sample size should be as specified in the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples of first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 First article sample waiver. A first article sample submitted and approved on a recent contract (a contract remaining in production or a contract out of production less than one year) may be accepted by the procuring activity in lieu of an additional first article inspection. When the first article sample is waived (see 6.2.1), the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved first article sample shall determine the minimum requirements of the current contract or order.

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6.5 Test facility for first article inspection. The Naval Ordnance Station, Indian Head, Maryland is the design agent for the product covered by this specification. As such, they are equipped to perform the first article inspection and static-firing tests specified herein. Except when special conditions warrant otherwise, the Naval Ordnance Station, Indian Head, Maryland should be designated as the test facility for the first article inspection and the production lot acceptance test firings.

6.6 Hazard information.

6.6.1 Loaded assembly hazard information.

Explosive Hazard Classification: Class 1, Division 4
Storage Compatibility: Group G
Department of Transportation (DOT) Hazard Class: Class C
DOT Marking: Explosive Power Device Class C
Handle Carefully - (Keep Fire Away)

6.6.2 Hazard notice. The igniter assembly described herein and certain of its components are flammable or explosive, or both, and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.7 Subject term (key word) listing:

Aircrew emergency escape system
Igniter assembly
Rocket Motor M119

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - OS
Air Force 99

Preparing activity:
Navy - OS

Project 1377-O903

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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